

Pinsara Weerasinghe

he/him/his • Colombo, Sri Lanka (Open to Relocation) • +94719596692 • pinsarakweerasinghe@gmail.com •
linkedin.com/in/pinsara • github.com/PinsaraKWeerasinghe • pinsara.com

Senior Backend Engineer | Java & AWS Cloud Microservices

University of Moratuwa Computer Science and Engineering Honors graduate with 6 years of experience architecting high-performance backend systems across the Fintech and Aviation domains. Spent 2.5 years in Fintech delivering low-latency Java and Spring Boot backend infrastructure to optimize transactional data processing. Advanced into Aviation tech for 3.5 years, designing cloud-native microservices and enterprise data hubs on AWS utilizing the Serverless Framework and AWS CDK. Combines deep server-side logic mastery with automated cloud provisioning to deliver fault-tolerant, high-throughput microservice architectures. Seeking to leverage scalable system design and cloud engineering expertise as a Senior Backend Engineer.

WORK EXPERIENCE

Serverless Guru • Remote • 02/2023 - Present

AWS cloud consultancy specializing in enterprise serverless architecture and microservices. Assigned as a Cloud Engineer to deliver high-scale data hub for Air Canada.

Senior Software Engineer / Cloud Consultant • Full-time

- **Architected a reusable Python utility library** for AWS Glue ETL jobs, optimizing data transformation workflows and **reducing AWS cloud compute costs by 20%**.
- **Engineered a cost-effective CloudWatch alarm triggering mechanism** utilizing log metric filters, streamlining automated alerting and **reducing CloudWatch expenditure by 2%**.
- **Engineered a real-time event persistence engine** using Kafka Connect and AWS Athena, enabling instant event debugging, rapid failure recovery, and **significantly enhancing overall system availability and fault tolerance**.
- **Optimized enterprise data migration pipelines**, refining execution logic and **reducing total data migration processing time by 30%**.
- **Led the development of a real-time customer data ingestion platform**, utilizing AWS Lambda and Apache Kafka topics to seamlessly process over **75% of total enterprise customer event workloads**.
- **Integrated OpenSearch indexing for high-traffic API endpoints**, offloading query patterns from primary databases to **reduce database load by 40%** and unblock downstream database integrations.

GTN Technologies (Previously Global Market Technologies) • Colombo, Sri Lanka • 06/2019 - 02/2023

A global fintech leader and institutional trading infrastructure provider connecting banks, asset managers, and wealth platforms to over 90 global capital markets via advanced e-trading engines.

Software Engineer • Full-time

- Engineered real-time OHLC data extraction components to offloading high-concurrency traffic from legacy monolithic systems, reducing processing latency by 15%.
- **Architected real-time data dissemination engines** using **Java, Spring Boot, WebSockets, and TCP Sockets**, streaming concurrent market feeds to enterprise trading clients while **reducing overall data latency by 20%**.
- **Engineered value-added price data pipelines for FinHub integration**, enhancing market data extraction workflows and expanding real-time analytical capabilities for financial end-users.
- **Optimized database performance by wrapping Apache Solr instances** to handle non-critical API read queries, accelerating text-search indexing speeds and **reducing primary database load by 25%**.
- Mentored and guided software engineering interns in architecting an automated anomaly detection tool, elevating team engineering standards and accelerating internal diagnostic capabilities.

Intern Software Engineer • Full-time

- Engineered automated log analysis and anomaly detection tool for real-time market price backends, streaming live health telemetry to internal monitoring dashboards to streamline incident response and maintain high availability.

EDUCATION

Bachelor of Science (B.Sc. Hons) in Computer Science and Engineering

University of Moratuwa • Moratuwa, Sri Lanka • 09/2016 - 03/2021

* Completed intensive 4-year professional engineering curriculum specializing in advanced software architecture, data structures, cloud systems, and distributed computing databases.

G.C.E. Advanced Level Examination in Physical Science (Com. Mathematics, Physics, Chemistry)

Mo/Malwaththawala National School • Wellawaya, Sri Lanka • GPA: AAA • 01/2001 - 08/2014

* Achieved straight 'A' passes across all core technical subjects. Secured a highly competitive Z-Score of 2.3135, earning a District Rank of 2 out of thousands of candidates nationwide.

PROJECTS

Meridiano Ceylon – Tourism Booking Platform (www.meridianoceylon.com) • 04/2026 - 07/2026

Independent / Commercial Project

- **Technologies Used:** Next.js, React, TypeScript, Firebase, Tailwind CSS
- Architected a production Next.js travel platform with integrated Firebase cloud services—delivering dynamic tour package management, real-time booking enquiries, user authentication, and secure media asset storage.

Data Analytics & ML Platform • 05/2020 - 03/2021

University of Moratuwa (Final Year Research Project)

- **Technologies Used:** Apache Spark, NLP, Python, Machine Learning, Data Analytics
- Developed a final-year ML analytics framework using NLP and Apache Spark to process large-scale driver telemetry and feedback, computing real-time QoS scores and dynamic rankings.

MAS Holdings Quality Process Automation • 10/2020 - 12/2020

Fcode Labs

- **Technologies Used:** Flutter, Dart, Firebase
- Engineered a cross-platform Flutter/Dart app with Firebase integration to digitize garment evaluation workflows—automating scheduling, measurement logging, and photo capture—while developing a smart fashion tool to capture and visually compare clothing items.

VOLUNTEERING & LEADERSHIP

Department of Computer Science & Engineering, University of Moratuwa

Organizing Committee Member

Coordinated event logistics, technical showcases, and visitor engagement for Moratuwa's flagship CSE exhibition.

SKILLS

Cloud & Serverless Architecture: AWS (Lambda, CloudWatch, DynamoDB, Event-Driven Architecture, Glue, Microservices Architecture, OpenSearch, REST & GraphQL APIs, S3, Serverless Framework, SNS), SQS

Programming Languages: Dart, Java, JavaScript, Node.js, Python, SQL, TypeScript

Frameworks & Search Engines: Apache Solr, Express.js, Nest.js, Next.js, React, Spring Boot

Databases & Data Streaming: Amazon DynamoDB, Apache Kafka, Firebase, MySQL, PostgreSQL

DevOps, CI/CD & Tools: AWS CloudWatch Dashboards, Bamboo, Bitbucket, Docker, Git, Litestream

Personal Interests: Automotive Engineering & Mechanical Restorations, Travelling and Outdoor Exploration